
Please circle your Class Section: C01 (Dr. Athar) C02 (Mr. Negm)

Name: _____ Student ID: _____

ELEC ENG 3CL4 (Introduction to Control Systems)
Midterm Examination 2

Department of Electrical and Computer Engineering, McMaster University
Instructors: Dr. Shahrukh Athar (C01) and Mr. Ayman Negm (C02), Winter 2023
Wednesday, March 15, 2023, 6:30 PM to 8:00 PM (90 minutes)

Examination Conditions This test is conducted under standard examination conditions. In particular:

- You are not to communicate in any manner with any other student.
- You are not to look at the questions nor start writing until told to do so by an invigilator.
- When the invigilator announces that the test is over, you are to stop writing *immediately*.

The penalties for violating these or any other standard examination conditions will be severe, including a zero on the test and the commencement of formal disciplinary proceedings.

Instructions

- There are four (4) questions on this examination paper. Answer all four questions in the space provided in this exam paper. You may use the back of the pages if necessary.
- When given permission to start writing, you must first ensure that your copy of the examination paper is complete and bring any discrepancies to the attention of your exam invigilator.
- Write your name and student number on the front page of this examination paper and on any additional booklet that you include with it.
- Write your answers neatly and in a logical fashion. We may refuse to mark answers which are difficult to decipher.
- Direct answers without proper justifications will lead to reduced grade. However, justifications should be concise and to-the-point.
- You may write your test in pencil, but booklets written in pencil or with pages torn out will not be remarked.
- Closed-books, closed-notes.
- Aids permitted:

One single-sided (8.5 x 11) crib sheet with only handwritten notes (Not printed/photocopied).

McMaster Standard Calculator (Casio FX-991MS/MS+).

- Total Marks: 80 (mark allocations for questions/parts are shown within questions).

Q1 (20)	Q2 (20)	Q3 (20)	Q4 (20)	Total (80)

Useful Information

- The solutions to the quadratic equation $ax^2 + bx + c = 0$ are $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.
- In the function $f(t) = Ae^{-t/\tau}$ the term τ is sometimes referred to as the time constant.
- The following Laplace transform pairs may be of use. Here $u(t)$ is the unit step function

$$\begin{aligned}
 u(t) &\longleftrightarrow \frac{1}{s} \\
 e^{-at}u(t) &\longleftrightarrow \frac{1}{s+a} \\
 \sin(\omega t)u(t) &\longleftrightarrow \frac{\omega}{s^2 + \omega^2} \\
 \cos(\omega t)u(t) &\longleftrightarrow \frac{s}{s^2 + \omega^2} \\
 e^{-at} \cos(\omega t)u(t) &\longleftrightarrow \frac{s+a}{(s+a)^2 + \omega^2} \\
 e^{-at} \sin(\omega t)u(t) &\longleftrightarrow \frac{\omega}{(s+a)^2 + \omega^2} \\
 t^n u(t) &\longleftrightarrow \frac{n!}{s^{n+1}} \\
 \frac{df(t)}{dt} &\longleftrightarrow sF(s) - f(0^-) \\
 \frac{d^2f(t)}{dt^2} &\longleftrightarrow s^2F(s) - sf(0^-) - f'(0^-) \\
 \int_{0^-}^t f(\lambda) d\lambda &\longleftrightarrow \frac{F(s)}{s}
 \end{aligned}$$

- A generic form for a rational Laplace Transform is $G(s) = K_G \frac{\prod_{i=1}^M (s+z_i)}{\prod_{j=1}^n (s+p_j)}$, where the set of finite zeros is $\{-z_i\}_{i=1}^M$ and the set of poles is $\{-p_j\}_{j=1}^n$.
- The final value theorem states that for a Laplace Transform $Y(s)$ with all its poles in the left half plane, except for at most one pole at the origin, $\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s)$.
- For a standard second-order system with a transfer function of the form $T(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$, when $\zeta < 1$, the poles are at positions $-\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$. The step response of such a system has the following properties:

- $y_{\text{step}}(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_n\sqrt{1-\zeta^2}t + \theta)$, where $\theta = \cos^{-1}(\zeta)$.
- 2% settling time, $T_s \approx \frac{4}{\zeta\omega_n}$.
- Percentage (maximum) overshoot, $P.O. = 100e^{-\pi\zeta/\sqrt{1-\zeta^2}}$. Hence, $\zeta = \frac{-\ln(P.O./100)}{\sqrt{\pi^2 + \ln^2(P.O./100)}}$.
- Time of peak, $T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}$.
- For $0.3 \leq \zeta \leq 0.8$, the 10%-90% rise time is $T_r \approx \frac{2.16\zeta + 0.6}{\omega_n} \approx \frac{2}{\omega_n}$.

- Steady state error constants:
 - Position error constant: $K_{\text{posn}} = \lim_{s \rightarrow 0} G_c(s)G(s)$
 - Velocity error constant: $K_v = \lim_{s \rightarrow 0} sG_c(s)G(s)$
 - Acceleration error constant: $K_a = \lim_{s \rightarrow 0} s^2G_c(s)G(s)$

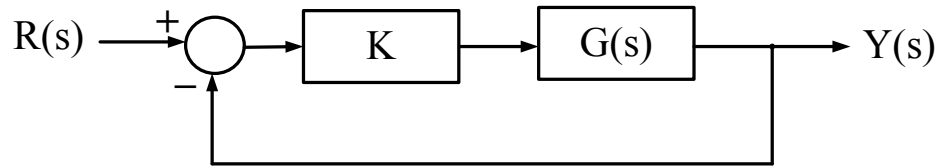
Question 1 (20 marks). Consider a system with the given closed-loop characteristic equation:

$$s^4 + 2s^3 + 5s^2 + 8s + K = 0$$

- a) **(8 marks)** Construct the complete Routh-Hurwitz Table for this system.
- b) **(3 marks)** For $K = 10$, how many poles are in the right-half s -plane?
- c) **(5 marks)** Find the range of K for which the system is stable.
- d) **(4 marks)** What value of K puts a pair of closed-loop poles on the $j\omega$ -axis, what is the position of the $j\omega$ -axis poles?

[Additional space for Question 1.]

Question 2 (20 marks).



Consider the closed loop feedback control system above, in which

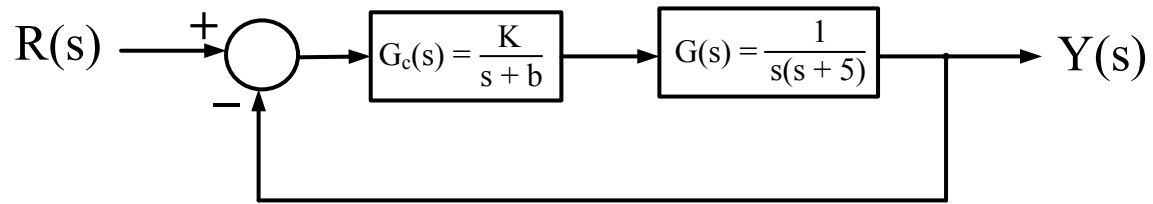
$$G(s) = \frac{(s+1)(s+7)}{(s-3)(s+3)(s+4)(s+5)}$$

For this system

- a) **(5 marks)** State the location of the open loop poles and zeros on the *s-plane*.
- b) **(5 marks)** For the Root Locus of this system, find the asymptotes, i.e., the number of asymptotes, their angles, and the centroid.
- c) **(7 marks)** Sketch the root locus for values of K from 0 to ∞ .
- d) **(3 marks)** Just by looking at the Root Locus that you drew, i.e., without doing any calculations, comment on whether this closed loop system becomes stable for any value of K .

[Additional space for Question 2.]

Question 3 (20 marks).

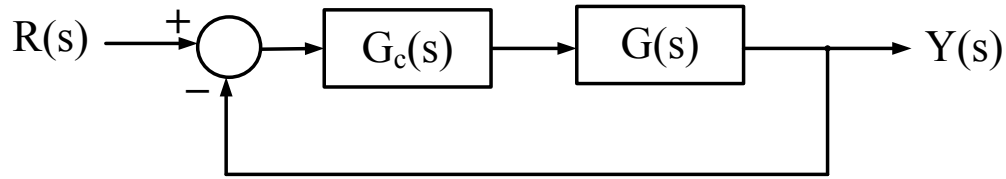


Consider the closed loop feedback control system above.

- a) (4 marks) Derive the transfer function of the closed-loop system, $\frac{Y(s)}{R(s)}$.
- b) (8 marks) Using Routh-Hurwitz criterion, state all the conditions on the controller parameters K and b to have a stable closed-loop system.
- c) (3 marks) State the condition on the controller parameters K and b to have a steady-state error of less than 10% for a unit-ramp input.
- d) (5 marks) Select a set of values for K and b , of your choice, that satisfies the conditions derived in Parts (b) and (c).

[Additional space for Question 3.]

Question 4 (20 marks).



Consider the closed loop feedback control system given in the figure above, in which

$$G_c(s) = \frac{K_c(s + a)}{s} \quad \text{and} \quad G(s) = \frac{1}{(s + 1)(s + 6)}$$

The step response of the closed-loop system should have a percentage overshoot that is at most 10% and a 2% settling time that is at most 2 seconds. The Root Locus approach will be used to find the controller parameters that fulfill the design requirements.

- a) **(5 marks)** Using the information above, find the location of the desired dominant closed loop poles.
- b) **(5 marks)** Find the acceptable range of the controller parameter a . [Hint: Make the Root Locus pass by the closed loop poles calculated in Part (a).]
- c) **(5 marks)** From the range of a found in Part (b), select a value of your choice and use it to sketch the Root Locus as K_c changes from 0 to ∞ .
- d) **(5 marks)** Find the value of the controller gain K_c that places the closed loop poles at the locations determined in Part (a).

[Additional space for Question 4.]